

source operating system like Ubuntu running on a machine available for retail is a fantastic thing to see. It is excellent because it helps people recognise it. Kudos to them and I wish them success. This will create awareness about open source technology, overall.

Q Does Fedora have any similar plans?

Q Fedora doesn't actually go and engage with vendors in that sense. So if someone wants to take Fedora and bundle it into the hardware that they sell, we are fine with that.

Q Can you share your insights on what the contribution of Indian developers is to the Fedora project?

Actually, we do have a significant number of developers in India. A lot of the work they do has to do with language support, like that of the Indian languages. We have people who contribute in terms of testing, and contributors who do work related to virtualisation. Most of them are Red Hat employees. Interestingly, this is how Red Hat hires. Most of the people we bring on board are because of their involvement in the community. That is a sure shot way of validating your skill set. So, even if a person is doing some localisation in the various Indian languages in Fedora, chances are that we may end up hiring him to continue to do the job that he has been doing anyway.

"If you look at the Fedora mailing list, you will find a lot of discussions, counter points and arguments on ways that things can be done in the project. A bulk of those commenting there are community people, who look at Fedora as their own project and there is a sense of personal ownership in the project as well."

Q What does Red Hat do to increase the involvement of developers in the Fedora project?

FUDCON is one of the main events that we use for the Fedora community. There is another event called the Fedora Activity Day, which comprises smaller Fedora conferences that are done over one day, mostly in the universities and institutes of higher learning, where there is a hackathon or a similar event relating to usage of technology. We have seen very good participation from the community in these events. There are a lot of other activities going on in different parts of India, like those in the IITs and other engineering colleges. 

Continued from page 83...

```
$ maxima -q
(%i1) string(solve(a*x^3 + b*x^2 + c*x + d = 0, x));
(%o1) [x = (-sqrt(3)*%i/2-1/2)*(sqrt(27*a^2*d^2+4*b^3-18*a*b*c)*d+4*a*c^3-b^2*c^2)/(2^3*(3/2)*a^2)-(27*a^2*d-9*a*b*c+2*b^3)/(54*a^3)]^(1/3)-(-sqrt(3)*%i/2-1/2)*(3*a*c-b^2)/(9*a^2*(sqrt(27*a^2*d^2+4*b^3-18*a*b*c)*d+4*a*c^3-b^2*c^2)/(2^3*(3/2)*a^2)-(27*a^2*d-9*a*b*c+2*b^3)/(54*a^3)]^(1/3)-b/(3*a), x = (sqrt(27*a^2*d^2+4*b^3-18*a*b*c)*d+4*a*c^3-b^2*c^2)/(2^3*(3/2)*a^2)-(27*a^2*d-9*a*b*c+2*b^3)/(54*a^3)]^(1/3)-(-sqrt(3)*%i/2-1/2)*(3*a*c-b^2)/(9*a^2*(sqrt(27*a^2*d^2+4*b^3-18*a*b*c)*d+4*a*c^3-b^2*c^2)/(2^3*(3/2)*a^2)-(27*a^2*d-9*a*b*c+2*b^3)/(54*a^3)]^(1/3)-b/(3*a), x = (sqrt(27*a^2*d^2+4*b^3-18*a*b*c)*d+4*a*c^3-b^2*c^2)/(2^3*(3/2)*a^2)-(27*a^2*d-9*a*b*c+2*b^3)/(54*a^3)]^(1/3)-b/(3*a), x = (sqrt(27*a^2*d^2+4*b^3-18*a*b*c)*d+4*a*c^3-b^2*c^2)/(2^3*(3/2)*a^2)-(27*a^2*d-9*a*b*c+2*b^3)/(54*a^3)]^(1/3)-b/(3*a)]
(%i2) quit();
```

And that's the power of Maxima.

Plotting graphs

Along with the algebraic processing, Maxima also supports

graph plotting for the algebraic expressions, though we must specify the intervals to plot it. It can do 2-D as well as 3-D plots. The following code shows an example for each of these. And Figures 1 and 2 show the respective plots.

```
$ maxima -q
(%i1) plot2d([sin(x), atan(x), tanh(x)], [x, -5, 5])$
(%i2) plot3d(sin(abs(x) + abs(y))/(abs(x) + abs(y)), [x, -12, 12], [y, -12, 12])$
(%i3) quit();
```

After this first glimpse of what Maxima can do, we are now ready to move on to the details of specific topics, like expression simplification, polynomials, etc. 

By: Anil Kumar Pugalia

The author is a hobbyist and gold medalist from NIT Warangal and IISc Bangalore. Mathematics and knowledge-sharing are two of his many passions. He shares his experiments with Linux and embedded systems through his weekend workshops. For more about him, visit <http://sysplay.in> or contact him at email@sarika-pugs.com.